

Set - A

1. $Y_i | X_i \stackrel{\text{iid}}{\sim} N(\text{mean} = X_i, \text{variance} = 0.01)$

[dnorm(.,.) takes precision as the argument]

$$X_i | \mu, \tau \stackrel{\text{iid}}{\sim} N(\text{mean} = \mu, \text{variance} = \frac{1}{\tau})$$

$$\mu \sim N(\text{mean} = 0, \text{variance} = \frac{1}{0.0001} = 10^4)$$

$$\tau \sim \text{Gamma}(\text{shape} = 0.01, \text{rate} = 0.01)$$

[Only uniform representation with variance or precision is allowed]

2. burn length = 10000

post burn length (before thinning) = 20000

post burn length (after thinning) = 10000

samples used for inference = 10000

3. posterior $\Theta | \underline{Y} = \left[\left(\underset{\downarrow}{\underline{X}}, \underset{\sim}{\mu}, \underset{\sim}{\sigma} \right) | \underline{Y} \right]$

• $\dim(\text{posterior}) = n+2 \quad n \times 1 \quad 2 \times 1$

Line-17: `params = c("m $\mu$ ", "sigma")`

requirements for the target parameters we want to get the MCMC output for

[suppressing the output of $\underline{X} \ n \times 1$]

4. Time series $SD(\sigma) = \frac{\text{posterior SD for } \sigma}{\sqrt{ESS}}$

$$\Rightarrow ESS(\sigma) = \left(\frac{0.02127}{0.0002285} \right)^2 \quad (\text{Ans})$$

5. 95% equal tailed CIs for

$$\mu: = [1.9890, 2.0980]$$

$$\sigma: = [0.2192, 0.3015].$$

6. Check

25% quantile for $\mu = 2.025 > 2$ 50% quantile for $\mu = 2.0433 < 2.05$.
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$\therefore$ Let $I = [2, 2.05]$

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Justification

$$\supseteq I_{1/4} = [2.025, 2.0433]$$

$$\therefore P[I \ni \mu] \geq P[I_{1/4} \ni \mu] = 0.25 \quad (\text{Ans})$$

7. $\text{sigma-sq} = 1/\tau$

$$\text{params} = c("m\mu", "sigma-sq")$$

• To draw inference of $\mu + 3\sigma$, define

$$\text{lambda} = m\mu + 3/\sqrt{\tau} \cdot (\tau)$$

$$\text{params} = c("m\mu", "sigma-sq", "lambda")$$

8. Geweke Statistics. $|Z| > 1.96$: Chain may not show good convergence to stationary distⁿ. at 95% confidence.

Both $z_{\mu} = 0.5217$, $z_{\sigma} = 1.1945$ evidence of "good" convergence.

Gelman-Rubin diagnostics not possible due to $n_{\text{chain}} = 1$ (atleast ≥ 2 needed).

Set-B

1. $Y_i | X_i \stackrel{\text{iid}}{\sim} N(\text{mean} = X_i, \text{variance} = \frac{1}{400} = 0.0025)$

[dnorm(.,.) takes precision as the argument]

$$X_i | \mu, \tau \stackrel{\text{iid}}{\sim} N(\text{mean} = \mu, \text{variance} = \frac{1}{\tau})$$

$$\mu \sim N(\text{mean} = 0, \text{variance} = \frac{1}{0.01} = 100)$$

$$\tau \sim \text{Gamma}(\text{shape} = 0.001, \text{rate} = 0.001)$$

[Only uniform representation with variance or precision is allowed]

2. burn length = 20000

post burn length (before thinning) = 40000

post burn length (after thinning) = 10000

Samples used for inference = 10000

3. posterior $\Theta | \underline{Y} = \left[\left(\underset{\substack{\downarrow \\ n \times 1}}{\underline{X}}, \underset{\substack{\sim \\ 2 \times 1}}{\mu}, \sigma \right) | \underline{Y} \right]$

$$\dim(\text{posterior}) = n + 2$$

Line-17: params = c("m μ ", "sigma")

requirements for the target parameters we want to get the MCMC output for

[suppressing the output of $\underline{X}_{n \times 1}$]

$$4. \text{ Time series SD}(\sigma) = \frac{\text{posterior SD for } \sigma}{\sqrt{\text{ESS}}}$$

$$\Rightarrow \text{ESS}(\sigma) = \left(\frac{0.02063}{0.0002021} \right)^2 \quad (\text{Ans})$$

5. 95% equal tailed CIs for

$$\mu: = [1.9116, 2.0210]$$

$$\sigma: = [0.2344, 0.3147].$$

6. Check 25% quantile for $\sigma = 0.257 > 0.25$
75% quantile for $\sigma = 0.2847 < 0.3$

$$\therefore \text{Let } I = [0.25, 0.3]$$

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Justification

$$\supseteq I_{1/2} = [0.257, 0.2847]$$

$$\therefore P[I \ni \sigma] \geq P[I_{1/2} \ni \sigma] = 0.5. \quad (\text{Ans})$$

$$7. \sigma_{sq} = 1/\tau$$

$$\text{params} = c("m\mu", "sigma_sq")$$

• To draw inference of $\mu + 3\sigma$, define

$$\text{lambda} = m\mu + 3/\sqrt{\tau} \cdot (\tau)$$

$$\text{params} = c("m\mu", "sigma_sq", "lambda")$$

8. Geweke Statistics. $|Z| > 1.96$: Chain may not show good convergence to stationary distⁿ. at 95% confidence.

Both $z_{\mu} = -0.4128$, $z_{\mu} = 0.4076$ evidence of "good" convergence.

Gelman-Rubin diagnostics not possible due to $n_{\text{chain}} = 1$ (atleast ≥ 2 needed).